Egg segmentation and tracking: report

Honesty statement. No human pixel-level annotation exists. All 'ground truth' masks are SAM-derived pseudo-ground-truth (Task 1 output) and all egg counts, instance verdicts and video identity segments are AI-assisted (Claude Code visual review) and unverified by a human. Metrics are agreement with these references, not accuracy. Claude Code also wrote most of the code; see the AI disclosure at the end. Repository: https://github.com/NITISH-R-G/nosh-robotics-egg-segmentation-tracking . Tracking video (Google Drive): https://drive.google.com/file/d/1UfviG9mgcDgNzNUQBNJYFNFK-SVJR7Bu/view?usp=sharing (the previously saved T3b tracking results drawn on the original footage, source frames 240-999, tracker updates at 4 Hz with boxes held at 24 fps; tracking was not rerun to make it; see section 7).

1. Summary

	dev (88 imgs)	eval (22 imgs)
S1 (SAM + selector) exact egg count	0.830 (73/88)	0.864 (19/22)
classical exact egg count	0.477 (42/88)	0.545 (12/22)
classical instance F1 vs reviewed reference (dev)	0.416	not computed (no eval reference)
video (T3b, proxy): ID switches / gaps retained / complete-detection coverage	0 / 3 of 3 / 0.963	same clip, 1.000 on held-out frames

The pretrained route (S1) is much more reliable than the classical one on this dataset; the classical method is a documented weak baseline that fails whenever the background is as bright as the egg. The two methods are not ranked by pixel accuracy because no independent mask reference exists. S1 under-counts (0.159 of dev images, 0.136 of eval images) with one over-count in 88 dev images, and flags images where nothing passes its gates.

2. Data and leakage-aware split

110 images (Roboflow egg export, CC BY 4.0). Exact (SHA) and near-duplicate pairs were linked into groups; groups are split 80:20 with seed 42, giving 88 dev / 22 eval images with no group crossing (`data/split_manifest.csv`). A first split (v1) was contaminated because pilot-inspected images ended in eval; v1 is kept for history and v2 forces those groups to dev. Constraints, grouping decisions and a leakage report are in `data/` and `src/data/splitting.py`; the evaluation images were not inspected or run until both methods were frozen.

3. Task 1: pretrained segmentation (S1)

SAM ViT-B (official checkpoint, sha256 recorded in the frozen file) automatic mask generator, 32x32 points, run once for all 110 images; raw candidates are cached (14,400 candidates). A feature selector (area 1-60%, solidity >= 0.95, aspect <= 2, ellipse IoU >= 0.93, predicted IoU >= 0.9, stability >= 0.9, NMS) picks the egg masks; images with no selection are flagged for review, never guessed. A pre-registered set of selector experiments and a root-cause audit are in `experiments/`. An early freeze was withdrawn after display artefacts looked like false positives; the audit found a display defect, not a pipeline bug. A fallback using appearance descriptors and a box-prompted boundary refinement were tested on dev and rejected against pre-registered criteria. Outputs for all 110 images: `outputs/task1_sam_masks/` (15 are empty/flagged: 13 dev, 2 eval).

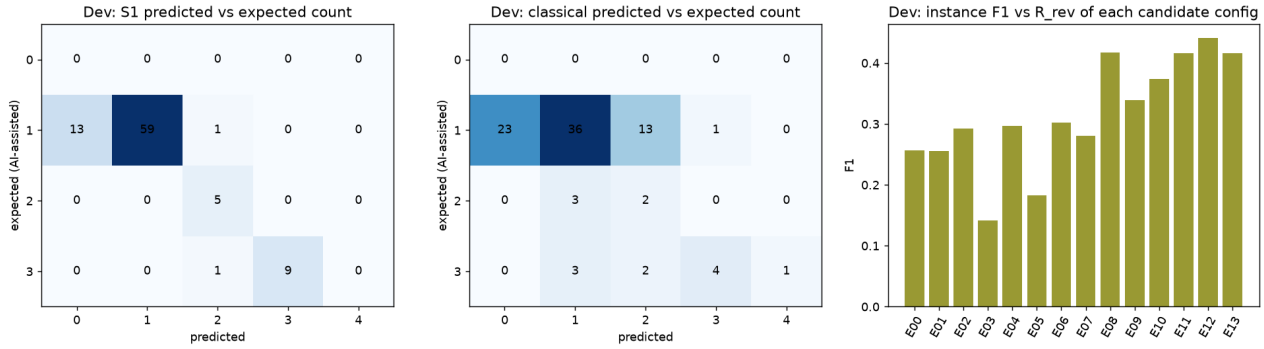
4. Task 2: classical segmentation and experiment log

Pre-registered 14 experiments (E00-E13), a greedy keep/discard rule evaluated on dev only, and leave-one-scene-out checks. Final config (frozen as classical_v1, config sha `6696e8cab3ba`): Lab whiteness (L minus chroma), Otsu, 5 px open and close, hole fill, area 1-60%, aspect <= 2.2, border-touching components >= 5% of the image. Every trial, including failures, is in `experiments/classical_log.csv` with full parameters; the table gives the change, why and the keep/discard result (deltas of dev metrics against the comparison base).

id	what changed / why	dev deltas vs comparison base (f1 vs R_rev, IoU vs R_rev, exact count)	decision
E00	baseline: simplest interpretable chain (Lab lightness, Otsu, 5 px open/close, fill holes, area 0.5-60%)	baseline	baseline
E01	HSV value as the foreground map: less affected by chroma than Lab L?	-0.002, +0.000, -0.045	discarded

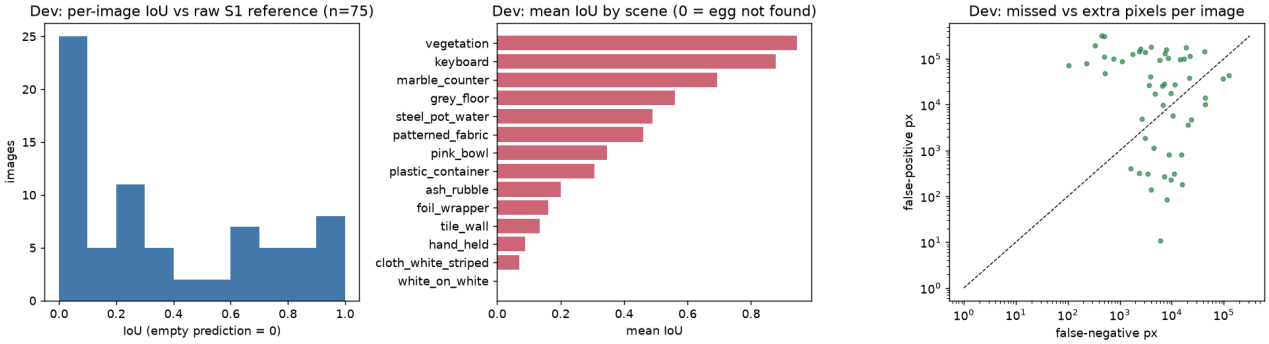
id	what changed / why	dev deltas vs comparison base (f1 vs R_rev, IoU vs R_rev, exact count)	decision
E02	whiteness (L - chroma): eggs are bright AND unsaturated; should reject coloured bright regions (cloth, bowls, tile)	+0.036, +0.026, +0.000	kept
E03	inverted saturation: eggs are nearly unsaturated	-0.115, -0.112, -0.011	discarded
E04	blur before Otsu to suppress texture (cloth, rubble) that splits regions	+0.005, -0.001, +0.057	discarded
E05	local threshold for uneven lighting and gradients	-0.110, -0.019, -0.273	discarded
E06	larger morphology to remove bridges to background and fill gaps	+0.009, +0.012, +0.057	discarded
E07	smaller morphology to keep thin egg boundaries	-0.012, -0.001, +0.000	discarded
E08	egg-like shapes are convex: reject ragged regions	+0.125, -0.030, +0.136	discarded
E09	eggs are compact: reject elongated bars	+0.046, +0.015, +0.045	kept
E10	reject larger specks and small background blobs	+0.035, +0.003, +0.023	kept
E11	border-touching components must be large (clipped eggs are large)	+0.043, +0.015, +0.136	kept
E12	separate touching eggs with a distance-transform watershed	+0.026, +0.009, -0.057	discarded
E13	shape prior: component must resemble its fitted ellipse	+0.001, -0.136, -0.193	discarded

Kept: E02, E09, E10, E11. Near misses decided by the rule: E08 (solidity) lost 0.030 macro IoU (limit 0.010); E12 (watershed) lost 0.057 exact-count rate (limit 0.030). Parameters of every trial: `experiments/classical_log.csv`.

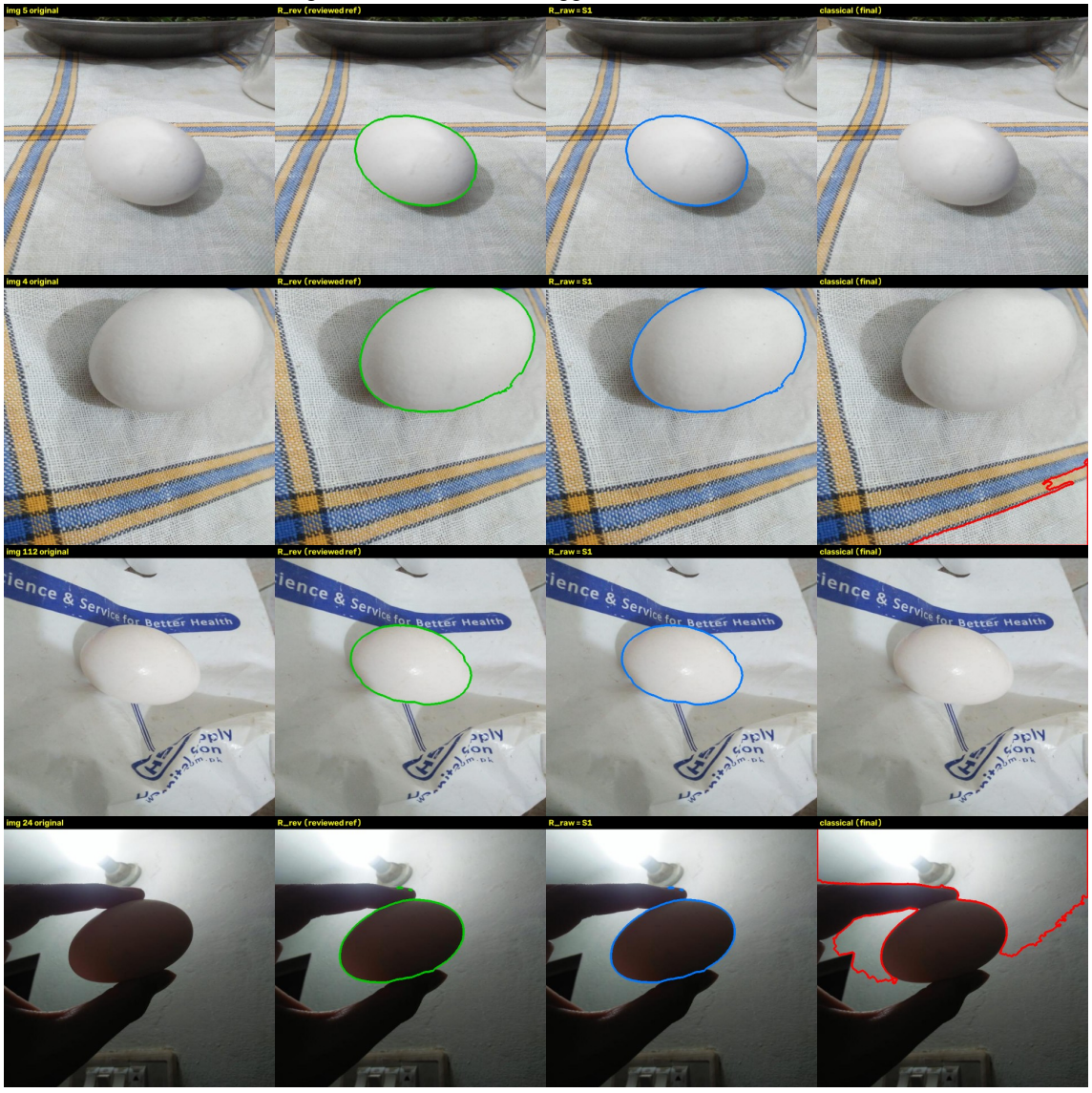


5. Task 3: evaluation and analysis (dev; Task 1 masks as reference)

Reference: S1 raw masks (R_raw) and S1 plus 7 AI-assisted additions (R_rev), kept separate. Dev, 88 images, classical vs R_rev: instance P/R/F1 0.442/0.393/0.416, macro IoU 0.363, macro Dice 0.441, count exact 0.477 (over 0.170, under 0.352); vs R_raw macro IoU 0.366. 20 images have the right count with wrong instances (false positive standing in for a miss) against 0 for S1. S1's reference-based numbers are not accuracy (the reference is S1); its independent measure is verdict precision 0.990 / recall 0.876. Per scene the classical F1 is 0 on ash, striped cloth, egg tray, foil, tile and white-on-white, and 1.0 only on a dark keyboard background. Runtime: classical 0.05 s/image CPU; SAM candidates 16.4 s/image GPU plus selector 0.26 s.



Qualitative: worst, best and category overlays (original | reference | S1 | classical). The failure is structural: brightness and low saturation describe the background as much as the egg.



Failure types (details: `experiments/classical_failure_analysis.md`): empty prediction on white-on-white, tile, cloth and pale bowls; backlit egg darker than the wall; touching eggs merged; bright reflective foil or water beats the egg;

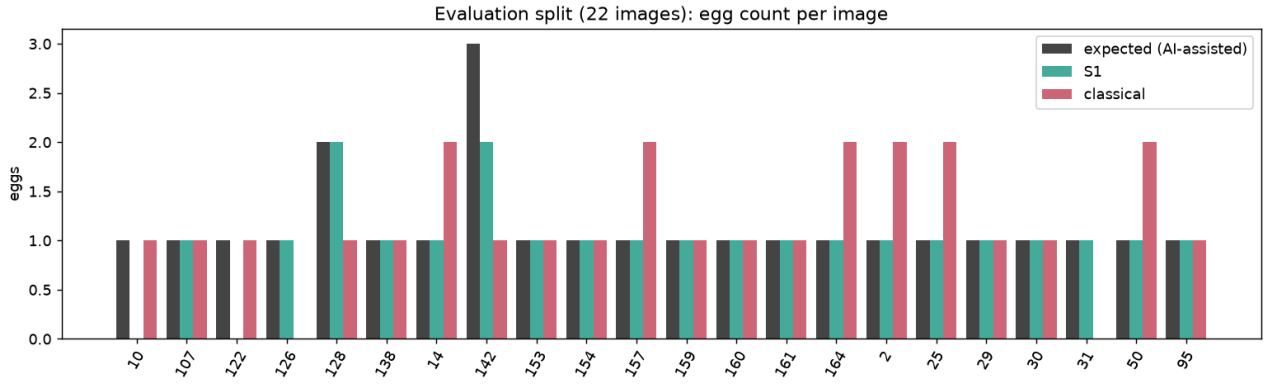
ash, skin and rug regions merged with the egg. Improvement directions (not implemented): learned or colour-texture models, a background model, per-image adaptive features; none fits the classical constraint cheaply.

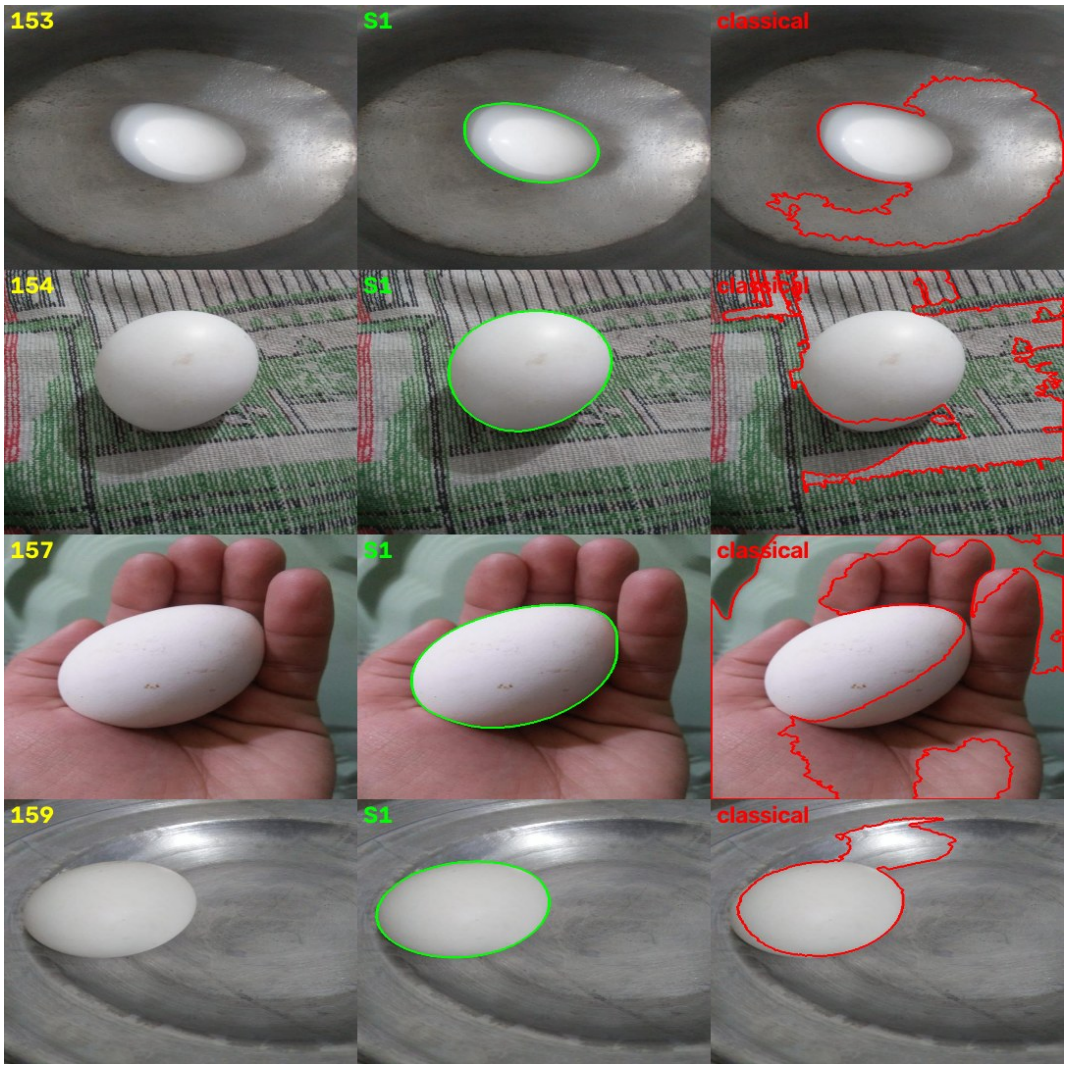
6. Frozen methods and final evaluation (22 untouched images, one run)

Both methods were frozen before the evaluation run with hashes, source-file hashes and dependency record (`configs/frozen/S1_v1.json` artifact sha `a6e46871795be95e`, `classical_v1.json` `d8c827fe7d9d12ff`); the run is recorded once in `outputs/eval/RUN_RECORD.json` and cannot be repeated. The protocol was written before freezing (`experiments/final_evaluation_protocol.md`). Eval counts were labelled by AI after the run from raw images but after the predicted counts had been printed (not blind). Note: `classical_log.csv` was appended with the freeze row after the freeze, so its current hash differs from the one recorded in classical_v1.json.

method	exact	over	under	mean abs err	predicted / expected eggs
S1	0.864 (19/22)	0.000	0.136	0.136	22 / 25
classical	0.545 (12/22)	0.273	0.182	0.500	26 / 25

Classical vs S1 raw masks (agreement only, S1 non-empty on 20/22): macro IoU 0.315, Dice 0.439; instances at IoU 0.5: TP 6, FP 20, FN 16. S1 misses: 10 and 122 (flagged), 142 (one of three touching eggs). Classical: empty on 126 and 31; background regions added or merged on most others. Dev and eval are consistent in direction; 22 images allow no significance claim.




153
S1
classical
154
S1
classical
157
S1
classical
159
S1
classical

7. Task 4: egg tracking in the video

Video Y4JyV94y2_A (AV1, 1920x1080, 24 fps); experiment interval frames 240-999 (title and explanation cards excluded). The frozen classical config also segments the wall, shirt and hand on this video, so a video-specific detector (grey >= 200, saturation <= 60, morphology, area/solidity/aspect/ellipse-fill filters; fixed a priori) is used. Sample: every 6th frame (4 Hz, 127 frames), alternate 1-second blocks as dev / held-out. Trackers: T0 (gated Hungarian), T1 (T0 + 'two eggs' prior) and, in a later improvement round, T2 (Kalman), T3 (Kalman + partial egg fragments) and **T3b (selected: T3 with prediction kept inside the frame)**. T0-T3 and their outputs are preserved unchanged.

	T0	T1	T2	T3	T3b (selected)
distinct IDs for 2 eggs	4	3	2	2	2
ID switches	3	2	0	0	0
gaps with identity retained (of 3)	0	1	3	3	3
complete-detection coverage dev / held-out	0.963 / 1.000	same	same	same	same
detector + tracker ms/frame (CPU)	8.3 + 0.115	8.3 + 0.106	6.1 + 0.18	9.7 + 0.22	9.8 + 0.21

What T3b adds. The pick-up failure of T0/T1 was mainly a detector problem: eggs under the hand were present as white fragments but rejected by whole-egg shape filters, so P had no detection from frame 732 to 810 and Q's detection was given P's track. T3b accepts fragments (max 3 per frame, area 0.3-5% of the frame, solidity >= 0.65, aspect <= 3.5) as *partial* observations that can update a track but never start one, predicts missing tracks with a constant-velocity Kalman filter (predicted boxes are reported separately and are never counted as detections), gates the Hungarian assignment on Mahalanobis distance and IoU and flags ambiguous assignments. T2 (Kalman without fragments) reaches the same headline numbers only because the long-lost track's growing uncertainty makes it a cheaper match at frame 750 (ambiguous, fragile): not selected.

Result and honest limits. On this single 31-second clip T3b has **0 identity switches and 3 of 3 annotated occlusion gaps retained** (T1: 2 and 1 of 3), measured against a single-annotator, AI-assisted proxy annotation (`data/video/identity_segments.csv`), not human ground truth. **This is not independent generalisation validation:** T3b was designed after inspecting the critical pick-up sequence (frames 690-780), the held-out block 744-767 had been viewed by the annotator and the designer, the annotation was written after viewing every sampled frame, and T3b was added after T3 failed a pre-registered plausibility criterion. Metrics saturate (0 / 3 of 3), so they cannot rank T3b against other good trackers. Scoring uses complete detections only (fragments are shown separately); detection coverage is unchanged: 0.963 dev (4 misses: caption arrow at 336/342, hand at 732/738) and 1.000 held-out, 0 extra complete detections.

Uncertainty is kept, not hidden. T3b flags 4 observed associations as uncertain (all 'ambiguous association'): frame 540 (ID 1 fragment re-acquired after the hand carry), 684 (ID 1 fragment re-acquired), 750 (ID 2 fragment during the pick-up) and 810 (ID 2 complete detection after the carry). ID 2 has no observation from frame 762 to 804: T3b shows a predicted box for 3 samples (762-774), then nothing (identity kept, position unknown) until the re-appearance at 810; the earlier T3 variant predicted up to frame 792 and its predicted centre left the frame (y < 0) at 780-792, which is why T3b bounds the prediction. Of 31 frames with fragments, all 21 fragments assigned to a track lie on an egg; 9 unassigned fragments are a false detection of the white 'Plane surface' caption arrow (frames 246-294), harmless because fragments cannot start tracks. Sensitivity (post-hoc, one factor at a time): T3b keeps 0 switches in 16 of 17 settings and fails at the lowest process noise (q_acc 0.1).

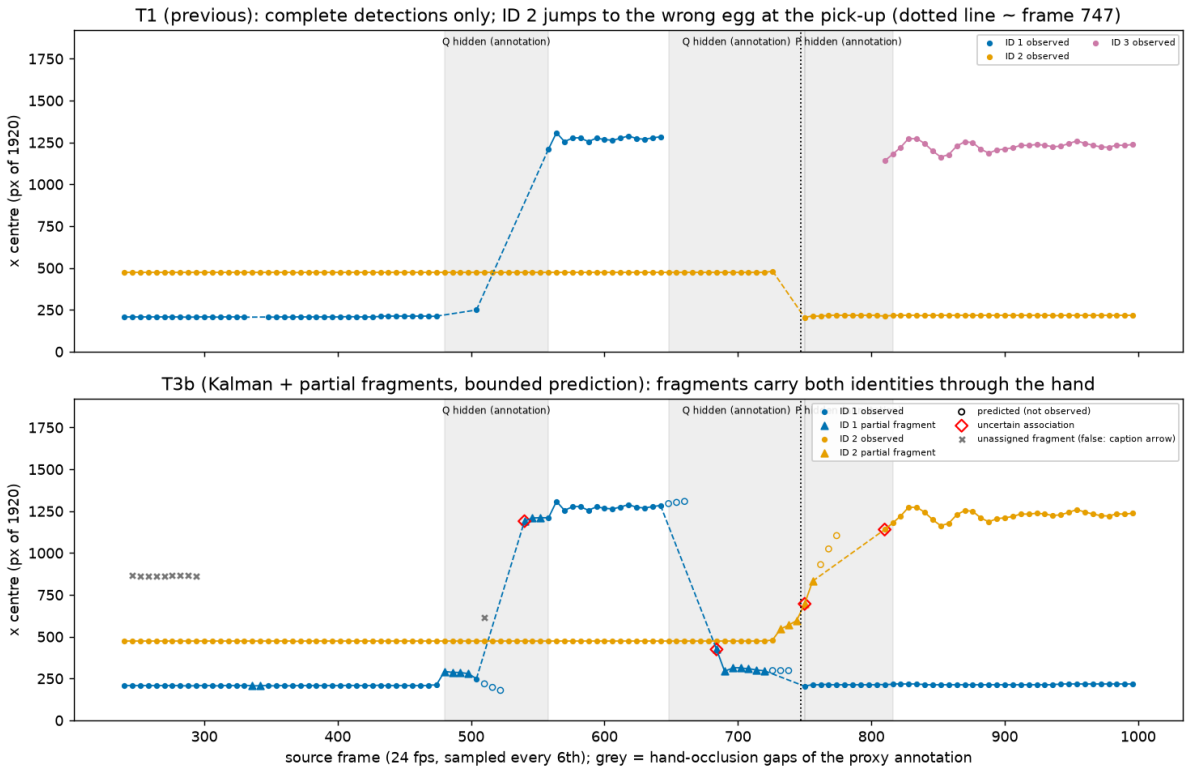


Figure: egg-box centre x per ID from the saved tracks (no third-party footage). Top T1: ID 2 jumps to the wrong egg at the pick-up (dotted line, about frame 747). Bottom T3b: dots = observed whole egg, triangles = fragments, hollow = predicted, red diamonds = uncertain association, grey crosses = false fragments; dashed lines = no observation between samples.

Task 4 output video: T3b tracking visualised on the original footage. Public Google Drive link: https://drive.google.com/file/d/1UfviG9mgcDgNzNUQBNJYFNFK-SVJR7Bu/view?usp=sharing. File `egg_tracking_T3b_footage_overlay_LOCAL_ONLY_v2.mp4` (the filename shown on Drive), 14,064,285 bytes, 1280x720, 24 fps, 31.67 s (760 frames), SHA-256 `9964d11a62d1c7f877ca1c38572fcf1487ded9c833c4419f54ff6c2eba982fa1`. The video draws the **previously saved T3b tracking results** (`outputs/video/tracking_T3b_kalman_partial_bounded/tracks.csv`) on the original MocomiKids footage. **Tracking was not rerun to produce it**: the renderer (`src/tracking/render_t3b_overlay.py`) reads only that CSV and the video, and every frame carries the caption 'T3b tracking output'.

Coverage and timing. The video covers **source frames 240 to 999 only** (760 of the clip's 1440 frames, 31.67 s of the 60 s clip). Earlier frames (title card and fade-in) and later frames (the transition and explanation card) are not shown and have no tracking output. The tracker updates at **4 Hz** (every 6th source frame, 127 updates); the boxes of each update are **held between updates at 24 fps**, so boxes lag the true motion in between (the caption shows the update frame and the hold offset).

Overlay in the video	Meaning
Solid box, colour = track ID (ID 1 green, ID 2 orange)	observed complete (whole-egg) detection
Dashed box in the ID colour	observed partial detection (fragment) assigned to that identity
Dotted box with yellow text 'PREDICTED, not observed'	predicted position (Kalman extrapolation); it is NOT a detection and is never counted as one
Grey dotted box	unassigned fragment (no track took it)
Red label ('? identity uncertain: ambiguous association')	uncertain identity association
Status panel (top right)	per ID: observed, predicted only, or 'NO ESTIMATE shown (not observed, no prediction)'

Limitations of the video (not hidden). (1) **Missing estimates:** at the update frames below an ID has no whole-egg observation. ID 1: fragments only at 336-342, 480-498, 540-552 and 684-720; predicted only at 510-522, 648-660 and 726-738; nothing at 528-534, 666-678 and 744. ID 2: fragments only at 732-756, predicted only at 762-774, nothing at 780-804 (the status panel says 'NO ESTIMATE shown'); an egg carried in the hand is therefore often unboxed. (2) **Uncertain identities:** four associations are flagged (frames 540, 684, 750 and 810, red labels); identity after the hand interaction rests on motion evidence and on a single-annotator, AI-assisted proxy annotation, not on human ground truth. (3) **Predicted positions can be inaccurate:** they are model extrapolations (for example the predicted boxes at source

frames 522 and 768 do not lie on the egg) and may be partly covered by the legend. (4) **Partial coverage:** frames 240-999 only. (5) Grey unassigned fragments at frames 246-294 are the on-screen 'Plane surface' arrow (a false fragment detection that cannot start a track); one more appears at frame 510. (6) This is a visualisation of proxy-evaluated results from one 31-second clip. It does **not** show continuous tracking of every egg in every frame, perfect identity preservation, or complete coverage of the video, and it is not independent validation. (7) The video contains third-party footage (MocomiKids, standard YouTube licence) and is provided to assess this assignment only; the original unannotated video is not redistributed. A footage-free schematic of the same saved tracks (`egg_tracking_T3b_schematic.mp4`) exists locally and is not uploaded. Details: `experiments/tracking_improvement_audit.md`, `experiments/tracking_improvement_results.md`, `experiments/video_tracking_results.md`.

8. Limitations and open issues

No human-verified labels; references derive from S1; 22 evaluation images; eval counts not blind; video evaluation is a proxy on one clip with 3 occlusion events; the Drive video shows saved results only (frames 240-999, 4 Hz updates held at 24 fps, intervals without estimates, four uncertain identity associations); the classical method is a weak baseline; S1 boundaries look smoother/larger than the visible egg edge (unmeasured). Four reviewed additions are first-pass only and seven development images have a bad-boundary S1 candidate excluded from the reviewed reference (one of them uncertain).

9. Reproduce

See README.md (setup, data/weights locations, exact commands). Tests: 213 pass. Evaluation re-run is blocked by design.

10. AI-tool disclosure

Claude Code (Anthropic) wrote most code, tests, protocols and analyses under the author's direction and produced the AI-assisted labels by viewing images; these labels are unverified by a human. The author is expected to explain the methods in the interview; withdrawn or rejected steps (premature selector freeze, contaminated split v1, rejected fallback and boundary refinement) are recorded in the logs.